



TEC initiative on AI Fairness Assessment Framework



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About biases and fairness in AI/ ML Systems

Need for independent audits for assessing AI fairness

Policy framework and Government initiatives

About the TEC initiative

Summary of responses received

Proposed action plan

Artificial intelligence (AI) and Machine Learning (ML) applications are increasingly being used in all domains.

- Healthcare, Agriculture, Smart Cities, Smart Homes, Finance, Defence, Transport, Logistics, Natural Language Processing, Surveillance...

AI/ ML Systems are also being used by Governments for delivery of public services and e-governance.

These systems use various algorithms to learn from training datasets, which could be real-life or synthetic data.

As these systems are deployed, they keep learning from their experience.

Biases can enter into these systems due to: (i) Imperfect training datasets, (ii) Algorithms used for learning and predictions, (iii) User interactions.

The biases could be on demographic attributes such as age, gender, marital status, race, caste, religion, nationality, etc. or on other attributes such as temperature, time, etc.

Various researchers have demonstrated that many AI Systems deployed by leading organisations and governments were biased and favoured a particular demography or discriminated against some demography. Some examples are cited as follows:

- An online AI-based recruitment system developed by Amazon was biased in favour of male candidates for technical jobs.
- Various facial recognition systems have biases and give accurate detection results only for people of a certain race.
- The US healthcare system preferred treating white patients.
- Online search queries are often biased in favour of certain companies and/ or products.
- A speech recognition system trained on US adults could not recognize evolving slang words or phrases when used by teenagers.

Bias in AI Systems raises various ethical, social and legal issues.

The users expect these systems to be fair and just in their outcomes.

A biased AI System prefers a certain demography while discriminating against others.

This causes imbalances in the society.

When used for e-governance or by the judiciary, their fairness becomes a legal requirement.

Every use case would have some harmful impact if the AI System used is not fair.

- A biased AI System used for Clinical Decision Support might lead to incorrect diagnosis, thus wrong treatment and/ or more visits to clinic.
- A biased Recommender System could harm the business prospects of startups or small enterprises.
- A biased IoT Based Preventive Maintenance System could trigger wrong alarms thus increasing maintenance costs.
- Bias in a driverless car could prove fatal for the passengers or the pedestrians.

The meaning of fairness may depend on the context in which it is applied. Different regions, social groups might perceive fairness differently.

One single definition of fairness would itself be unfair.

On a broad level, a fair AI System is expected to treat two different instances similarly if both differ only on protected attributes and are otherwise similarly placed.

On the other hand, a biased system will treat the two instances differently on the basis of the difference in their protected attributes.

A real-life AI System would have multiple protected attributes. A system might be fair with respect to some of these protected attributes and biased for some others.

It is therefore important to check for fairness for all the protected attributes in the application and not just one or a few of them.

Presently, assessing fairness of AI Systems is not standardised. Some developers carry out some checks; many developers may not even be aware of the concept of fairness.

The users, especially government organisations, also do not have any mechanism to know whether the AI Systems they are deploying are fair and unbiased.

While there are proper procedures for third-party audits and certification of websites and portals, there is no such established procedure for assessing fairness of AI Systems.

In view of the potential harmful effects of unfair applications, third-party assessment for fairness of AI Systems may be required, especially when public good is involved.

This involves (i) standardising the process for evaluating fairness of AI Systems, and (ii) their independent assessment and certification as per the standardised procedure.

National Digital Communications Policy 2018 - enumerates the role of Dept. of Telecom in the field of AI:

2.2 Ensuring a holistic and harmonised approach for harnessing Emerging Technologies

- (a) **Synergising deployment and adoption of new and emerging technologies** by:
 - i. Creating a roadmap for emerging technologies and its use in the communications sector, such as 5G, **Artificial Intelligence**, Robotics, Internet of Things, Cloud Computing and M2M.
- (g) **Leveraging Artificial Intelligence** and Big Data in a synchronized and effective manner to enhance the overall quality of service, spectrum management, network security and reliability.

2.3 Research and Development

- (a) Promoting research & development in Digital Communication Technologies by:
 - iii. **Creating a framework for testing and certification of new products and services**

Niti Aayog - National Strategy for Artificial Intelligence

National Strategy for Artificial Intelligence #AIforAll, June 2018.

- Focuses on how India can leverage the transformative technologies to ensure social and inclusive growth in line with the development philosophy of the government.
- India's approach to implementation of AI has to be guided by optimisation of social goods, rather than maximisation of topline growth.
- **Underlines the importance of a trusted ecosystem for accelerated adoption of the technology.**

Approach Document for India Part 1 – Principles for Responsible AI, February 2021

- This approach paper aims to establish broad ethics principles for design, development and deployment of AI in India
- An essential roadmap for the AI ecosystem, encouraging **adoption of AI in a responsible manner in India** and **building public trust in the use of this technology**, placing the idea of 'AI for All' at its very core.

Approach Document for India: Part 2 - Operationalizing Principles for Responsible AI, August 2021

- This paper identifies a **series of actions** that the **ecosystem** must adopt to drive responsible AI.
- These actions are divided among three stakeholders; governments, the private sector and research institutions.

For the government – designing ideal regulatory and policy interventions, creating awareness, accessibility and capacity building, and facilitating precise procurement strategies.

For the private sector and research institutions – incentivising ethics by design, **creating frameworks** for compliance with relevant AI standards and guidelines, and the **promotion of Responsible AI practices** in research.

- fill the gap regarding fairness assessment of AI systems
- facilitate startups & MSMEs
- implement the mandate of NDCP-2018
- complement NITI Aayog's National Strategy for AI

Framing a Standard
for assessing
fairness of AI
Systems

Assessing data-
driven AI Systems
for fairness/ bias **on
voluntary basis**

Issuing fairness
rating/ certification
as a benchmark of
fairness

Fairness Certification proposed by TEC is going to address a vital ethical and social issue.

Also going to meet the legal requirements of future.

As AI systems are being used in almost all aspects of our lives and even e-Governance applications would be using Artificial Intelligence, the initiative of TEC would benefit every segment of the society.

While procuring AI-based applications through a transparent tendering process, Fairness Certificate could be a qualification criterion.

Developers would like to get their systems certified for fairness from third-party auditors for the wider acceptability of their products.

Start-ups, MSMEs and even large enterprises, Researchers, College students and other individuals, Government organisations, International organisations such as World Bank, etc. would be benefitted from the TEC initiative.

TEC initiated stakeholder consultations vide letter no. 4-3/2022-C&B/TEC/AI-Fairness dated 22-02-2022.

Invited suggestions for framing procedures for assessing fairness for different types of AI/ ML Systems.

Last date for accepting inputs extended to 25-03-2022 (earlier 08-03-2022).

Held a webinar on 22-03-2022.

Inputs were sought on 9 questions.



Department of Telecommunications
Telecommunication Engineering Centre
K.L. Bhawan, Janpath, New Delhi – 110001
(Convergence & Broadcasting Division)



No.: 4-3/2022-C&B/TEC/AI-Fairness

Date: 22.02.2022

Subject: Framework for Fairness Assessment of Artificial Intelligence/ Machine Learning Systems by TEC - building public trust in AI/ ML Systems #AIforAll

Artificial intelligence (AI) and Machine Learning (ML) applications are increasingly being used in all domains such as Healthcare, Agriculture, Smart Cities, Smart Homes, Finance, Defence, Transport, Logistics, Natural Language Processing, Surveillance, and so on. Various Government Organisations are also using AI/ ML Systems for delivery of public services and e-governance.

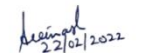
Researchers have shown that many such Systems deployed worldwide were having unintended biases. Bias in AI/ ML Systems raises various ethical, social and legal issues. While the users expect these systems to be fair in their outcomes, a biased AI/ ML System prefers a certain demography while discriminating against others. When AI/ ML Systems are used for e-governance or by the judiciary, checking for their fairness would become a legal requirement. Therefore, one important requirement of Responsible AI is that the AI/ ML Systems should be unbiased or fair.

National Digital Communications Policy-2018 mandates for synergising deployment and adoption of AI. National Strategy for Artificial Intelligence #AIforAll, and the Approach Documents for India, released by NITI Aayog, establish broad ethical principles for design, development and deployment of AI in India and provide an essential roadmap for the AI ecosystem, encouraging adoption of AI in a responsible manner in India and building public trust in the use of this technology, placing the idea of 'AI for All' at its very core.

With the aim to build public trust in AI/ ML Systems, TEC is working on Voluntary Fairness Assessment of AI/ ML Systems. As these systems are being used in all aspects of our lives, the initiative of TEC would benefit every citizen of the country. Startups and MSMEs in particular and even large enterprises will benefit as their products will be more credible and acceptable if they are assessed and certified by a neutral government agency such as TEC. It would also act as a benchmark for fairness, so Researchers, College students and other individuals, Government organisations and all segments of the society would benefit at large.

Accordingly, TEC is initiating stakeholder consultations and invites suggestions for framing procedures for assessing fairness for different types of AI/ ML Systems. Your valuable inputs on the matter as per the enclosed template may be emailed to avinash.70@gov.in and adic1.tec@gov.in by 08-03-2022 please.

This issues with the approval of Sr.DDG & Head TEC.


(Avinash Agarwal)

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Department of Telecom, New Delhi
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To:

1. All stakeholders

Copy for kind information to:

1. CEO Niti Aayog
2. Secretary Telecom/ Agriculture/ DST/ Defence/ Electronics & IT/ MoHUA/ MoRT&H
3. Member (Services)/ Member (Technology)/ DG (Telecom) DoT
4. Principal Scientific Adviser (PSA) to the Government of India
5. Chairman, TRAI/ Chairman DRDO
6. CEO Prasar Bharati
7. DG ICMR/ DG CDAC/ DG BIS
8. DG NTIPRIT
9. Directors of all IITs

For assessing fairness of various AI/ ML Systems, whether classification based on machine learning algorithms used is suitable or any other classification is suggested?

For assessing fairness of various supervised learning based AI/ ML Systems, whether a single assessment procedure is sufficient or further sub-classification is required?

What are the various types of unintended biases that should be taken into account while assessing fairness of various AI/ ML Systems?

Which fairness metrics/ parameters are appropriate for measuring bias/ assessing fairness of supervised learning AI/ ML Systems?

What could be the step-by-step procedure for assessing and rating a supervised learning AI/ ML Systems for fairness?

Whether any separate procedure is required for black-box testing of supervised learning AI/ ML Systems for fairness?

What are the best practices for handling code, training datasets, etc. by the third-party assessing agency?

Whether the fairness assessment procedure framed for supervised learning systems would also be applicable for assessing other types of AI/ ML Systems or different procedures need to be framed?

What are the additional considerations for framing fairness assessment procedures for various types of AI/ ML Systems?

TEC No.: 4-3/2022-C&B/TEC/AI-Fairness

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Framework for Fairness Assessment of Artificial Intelligence (AI)/ Machine Learning (ML) Systems by TEC

Template for submitting inputs
(Inputs may be provided as per the following template)

1. Types of AI/ ML Systems

There can be different ways to classify AI/ ML Systems. Based on the machine learning algorithms used, they can be classified as supervised, semi-supervised, unsupervised and reinforcement learning systems.

Question 1: For assessing fairness of various AI/ ML Systems, whether classification based on machine learning algorithms used is suitable or any other classification is suggested?

2. Supervised learning systems

Supervised learning systems learn from labelled datasets and are used to classify data or predict outcomes of unforeseen data accurately. They are typically used for various predictions and forecasting, face detection, handwriting/ signature recognition, etc. Some applications could include identifying fraudulent benefits claims, CCTV surveillance, social media sentiment analysis, spam detection, weather forecasting, stock price predictions.

Question 2: For assessing fairness of various supervised learning based AI/ ML Systems, whether a single assessment procedure is sufficient or further sub-classification is required?

3. Types of possible Biases in supervised learning systems

An AI/ ML application could have different types of biases. Also, some type of biases might be typical to the types of AI/ ML Systems. Some types of bias are selection bias, measurement bias, recall bias, observer bias, exclusion bias. The biases could be due to datasets used for training, algorithms and/ or usage of the System.

Question 3: What are the various types of unintended biases that should be taken into account while assessing fairness of various AI/ ML Systems?

4. Fairness Metrics/ parameters

Fairness of various AI/ ML Systems is usually measured using various metrics. These metrics provide a mathematical definition of fairness. Some commonly used fairness metrics are demographic parity, equal opportunity, equal mis-opportunity, average odds. Many fairness metrics are mutually exclusive;

Question 4: Which fairness metrics/ parameters are appropriate for measuring bias/ assessing fairness of supervised learning AI/ ML Systems?

5. Standardising procedure for fairness assessment of supervised learning AI/ ML Systems

While developers may test their AI/ ML applications using various fairness metrics and their own internal processes, there is a need to standardise the assessment process. The standardised process could be used by the developers for self-assessment as well as by third-party auditors for assessment and certification.

Question 5: What could be the step-by-step procedure for assessing and rating a supervised learning AI/ ML Systems for fairness?

6. White-box vs Black-box testing

Ideally, the assessing agency may expect access to the training datasets and the implementation details of the AI/ ML System. This may be considered as White-box testing. There might however be situations when the training datasets and/ or implementation details are not available with the assessing agency. This may be considered as Black-box testing.

Question 6: Whether any separate procedure is required for black-box testing of supervised learning AI/ ML Systems for fairness?

7. Procedures for handling code and training datasets by the assessing agency

When the assessment is to be carried out a third-party then it is important the standard procedures are in place for safe handling of sensitive material such as code, datasets, etc.

Question 7: What are the best practices for handling code, training datasets, etc. by the third-party assessing agency?

8. Semi-supervised, unsupervised and reinforcement learning and other type of systems

Besides supervised learning systems, other types of AI/ ML Systems could be Semi-supervised, unsupervised or reinforcement learning systems. Unlike supervised learning systems, the unsupervised learning systems use unlabelled data to identify patterns. Semi-supervised learning systems are a mix of the supervised and the unsupervised learning systems. Reinforcement learning generally learns from new situations using a trial-and-error method.

Question 8: Whether the fairness assessment procedure framed for supervised learning systems would also be applicable for assessing other types of AI/ ML Systems or different procedures need to be framed?

9. Any other inputs

The stakeholders may provide inputs on any other aspect that has not been covered in this template. These could include uncovered points, challenges foreseen, standard testing tools, etc.

Question 9: What are the additional considerations for framing fairness assessment procedures for various types of AI/ ML Systems?

IBM	Tata Consultancy Services Ltd	Reliance Jio Infocomm Ltd.	Salesforce	IIT Bombay	IIT Hyderabad
Smart Cities Mission MoHUA	Bureau of Indian Standards	CDAC	LDRA Technology Pvt. Ltd.	Analytics India Machine	Internet Freedom Foundation
Smartshift Logistics Solutions Pvt. Ltd.	Experian	Pradjna Intellisys Pvt Ltd.	Shri Ashish Airon, Member WHO-ITU	Sundar Narayanan, Researcher, AI & Tech Ethics	Asia Internet Coalition (AIC)
BSA The Software Alliance	Broadband India Forum (BIF)	Information Technology Industry Council (ITI)	IAMAI	NASSCOM	ASSOCHAM

For assessing fairness of various AI/ ML Systems, whether classification based on machine learning algorithms used is suitable or any other classification is suggested?

- Framework should take into account both the underlying technology and use case it is applied to.
- Use the output of machine learning algorithms with classification objectives for fairness assessment.
- The classification should be based on the associated risk with the AI system.
- ML algorithms offer a good perspective and may be sufficient for assessing fairness.
- Assessment of fairness may vary based on multiple broad factors such as the technology, application, or implementation.

For assessing fairness of various supervised learning based AI/ML Systems, whether a single assessment procedure is sufficient or further sub-classification is required?

- There are over 21 definitions of fairness.
- The framework should optimise for a definition with the least harm for all sub-populations.
- May require multiple assessment procedures for different application settings.
- The metric of fairness is application-specific.
- If there are multiple applications sharing the same metric, different parameters (e.g., thresholds) can be set that are appropriate for those applications.

What are the various types of unintended biases that should be taken into account while assessing fairness of various AI/ ML Systems?

- **Biases in Data Generation:** Historical Bias, Population Bias, Self-Selection Bias, Social Bias, Behavioural Bias, Temporal Bias.
- **Biases in Data:** Measurement Bias, Omitted Variable Bias, Representation Bias, Aggregation Bias, Sampling Bias, Longitudinal Data Fallacy, Linking Bias.
- **Biases in the Algorithm:** Algorithmic Bias, User Interaction Bias, Popularity Bias, Emergent Bias, Evaluation Bias.
- **Biases during Human Review:** Behavioural Bias, Presentation Bias, Content Production Bias, Social Bias.
- **Biases during Action:** Deployment Bias.

Which fairness metrics/ parameters are appropriate for measuring bias/ assessing fairness of supervised learning AI/ ML Systems?

- Different tools present different metrics
- The metrics used depend on the context of use
- It also depends on whether we intend to check the data or model or both
- Statistical parity
- Disparate impact
- Predictive parity
- Predictive equality for FPR
- Equal opportunity for FNR
- Equalized odds
- Overall accuracy equality
- Treatment equality
- Average odds difference
- TPR Diff
- Error rate difference
- Demographic Parity

What could be the step-by-step procedure for assessing and rating a supervised learning AI/ ML System for fairness?

- A combination of qualitative and quantitative procedures can be used to assess fairness of AI / ML models at different stage of AI pipeline
- This standardisation requires 2 things:
 - fulfillment of must-have/ mandatory conditions like regulatory, data privacy, security aspects.
 - Non Mandatory/ good to have aspects like modularity/ portability, repeatability, availability, scalability, outlier conditions.
- A well-crafted checklist at every stage of learning can help to identify potential entry of bias into the learning process.
- fairness checking is not a one-time procedure and it requires to be continuously checked while the model is deployed.

- Fairness has to be embedded into each step of the AI development lifecycle:
- **Design**
 - Gather domain-related inputs from practitioners.
- **Pre-processing**
 - Most bias is introduced during data acquisition and preparation. Assess your labels, categories, and data cleaning methods for bias.
- **In-processing**
 - The model must be assessed for accuracy, fairness, explainability, distributional robustness, and adversarial robustness.
- **Post-processing**
 - After the model is trained, check the group bias and individual bias on the model.
- **Post-launch**
 - to regularly monitor the AI system post launch not only to detect whether the pre-launch analysis was correct but also because new forms of bias can be introduced.

Whether any separate procedure is required for black-box testing of supervised learning AI/ ML Systems for fairness?

- In black-box testing, assessing agency will have access to the model and the test dataset only. Hence a separate procedure should be derived for black-box testing of supervised learning AI/ML systems for fairness.
- For black-box testing some techniques like equivalence partitioning and boundary value analysis are used.
- Interface standardization will be an initial challenge.
- Confidentiality and intellectual property related issues would pose a unique challenge in this regard as well.
- The focus should be on the process.
- White-box testing should be considered for self-certification by the developers themselves based on standard checklists.

What are the best practices for handling code, training datasets, etc. by the third-party assessing agency?

- Access to Code and training datasets should be logged/ recorded.
- Only (very few) authorized people should have direct access to code and data with record of access.
- Access (read) of the data only through APIs.
- Code and datasets should never be shared/ allowed to see by developers.
- Signing of the Non-Disclosure Agreement (NDA) before sharing any sensitive information.
- Regulatory policies defining all the security and privacy guidelines for AI systems categorized based on the sensitivity.
- A legal framework could be considered where, depending on the requirements and use case, the developers can be mandated to share their training datasets with the certifying agency for certification.

Whether the fairness assessment procedure framed for supervised learning systems would also be applicable for assessing other types of AI/ ML Systems or different procedures need to be framed?

- The fairness assessment procedures designed for one class AI/ML systems cannot be directly applied for other AI/ML systems, especially for Re-enforcement learning, Ranking problems.
- The procedure that is to be adopted would depend on the complexity of the algorithm, type of the algorithm, sensitivity of the use case, and stage of data science lifecycle.
- Testing can be used in a black-box model.

What are the additional considerations for framing fairness assessment procedures for various types of AI/ ML Systems?

- Collaborations and partnerships across stakeholder groups, such as industry and academia, to identify best practices and general principles that can be modified to specific use cases by developers and companies who use AI tools.
- Each organization should be empowered to self-establish processes to identify and mitigate bias in their use cases.
- Focus should also be on Explainability.
- Fairness assessment requires domain knowledge, application trade-offs that require specialized training, and knowledge related to fairness. Fairness education has to be strengthened to train the existing personas to achieve such specialized knowledge.

We need a standardised operating procedure as well as a standard score to evaluate and compare different AI Systems for fairness.

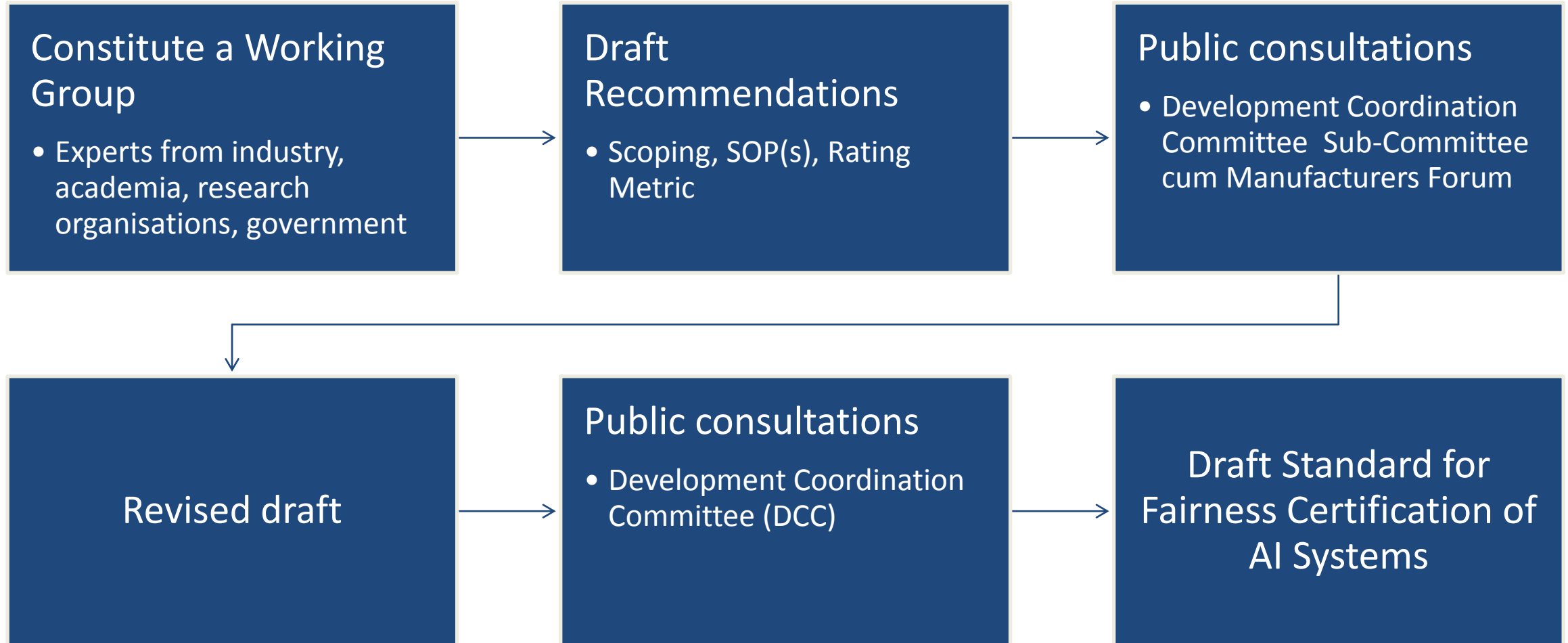
One SOP may not be sufficient for different situations or different types of AI Systems.

Various fairness metrics are already being used to measure AI fairness. We need a common score for comparison on top of these metrics.

Our paper proposes an **SOP** for Supervised Learning Systems and also introduces **Fairness Score** and **Bias Index** for comparison and rating [1].

It could serve as a background paper for reference.

- [1] Agarwal, A., Agarwal, H. & Agarwal, N. Fairness Score and process standardization: framework for fairness certification in artificial intelligence systems. *AI Ethics* (2022). <https://doi.org/10.1007/s43681-022-00147-7>



The voluntary Fairness Certification proposed by TEC is going to address a vital ethical and social issue.

Also going to meet the legal requirements of future.

Volunteers invited for the proposed Working Group.

TEC will follow a consultative process for framing the Standard.

The initiative of TEC would benefit every segment of the society and help in building public trust in AI/ ML Systems.

Thank You !